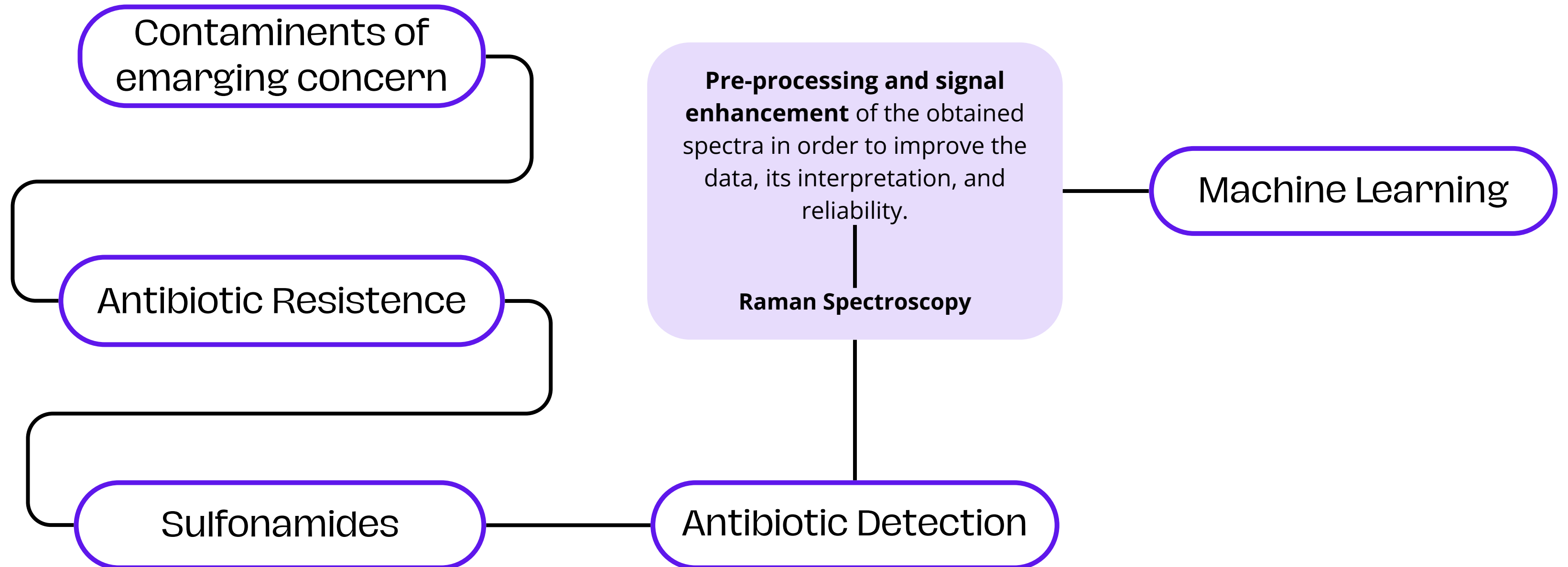




Improving Surface Enhanced Raman Spectroscopy (SERS) Preprocessing and Analysis

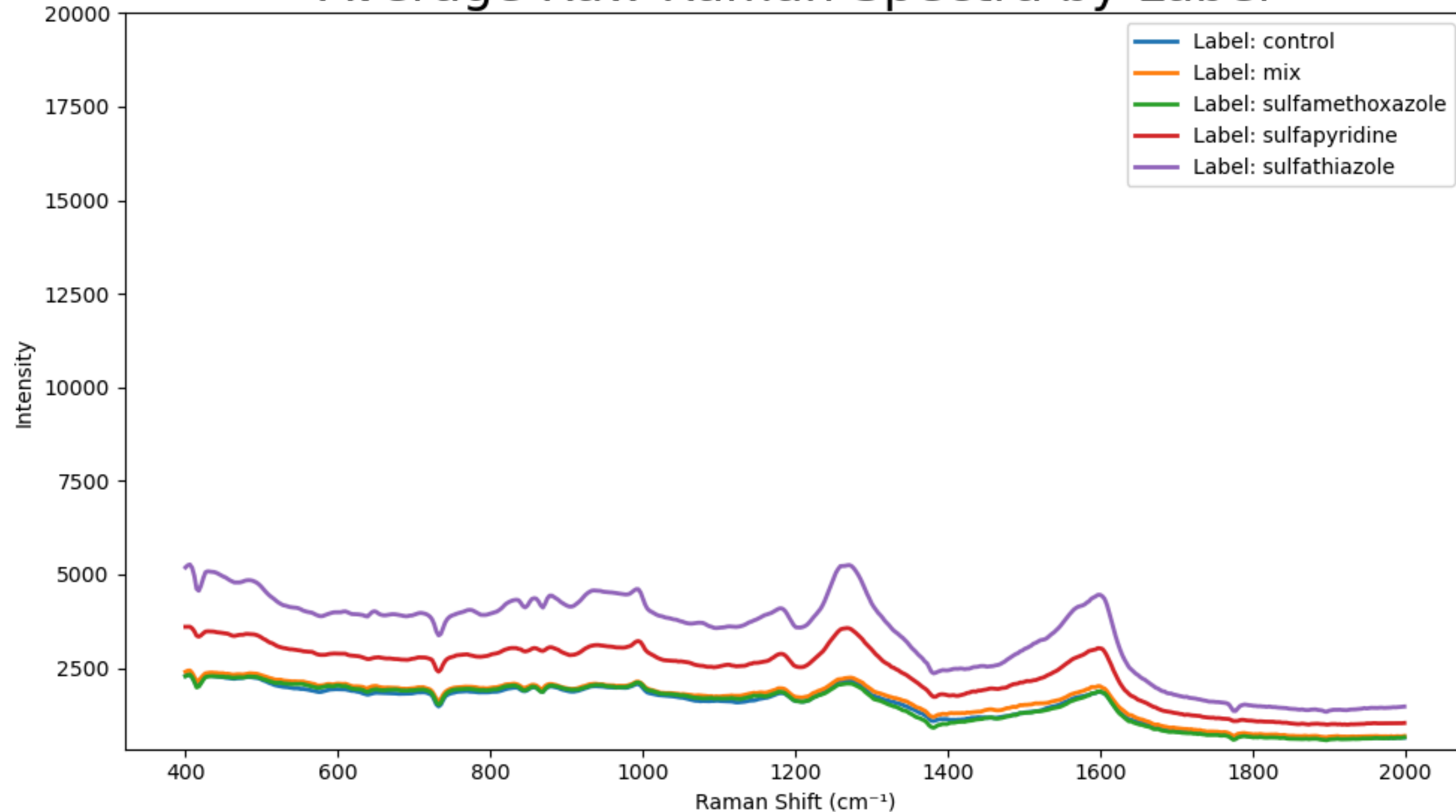
Bioinformatics MSc Project

Introduction



Surface-enhanced Raman Spectroscopy (SERS)

Average Raw Raman Spectra by Label

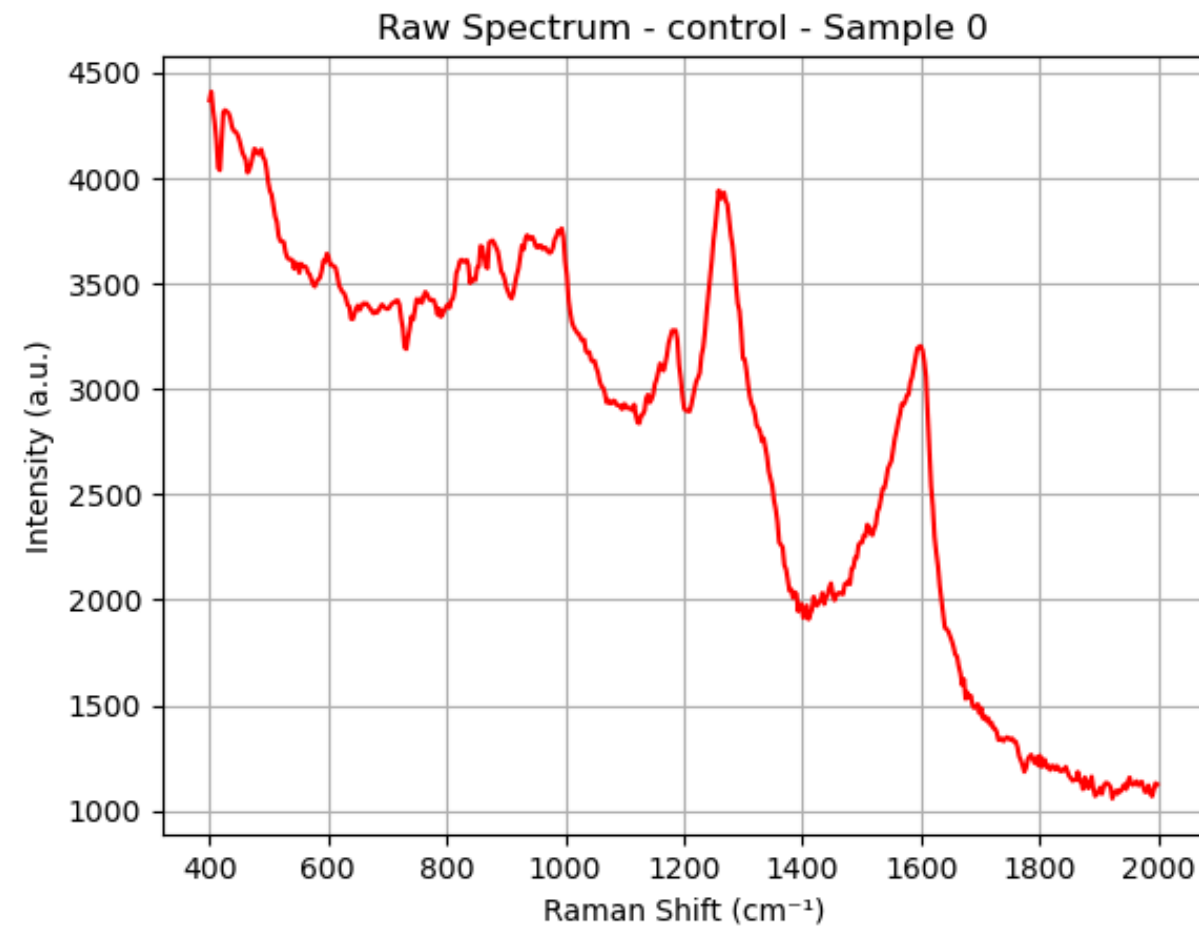


Total samples - 729

Samples per label:

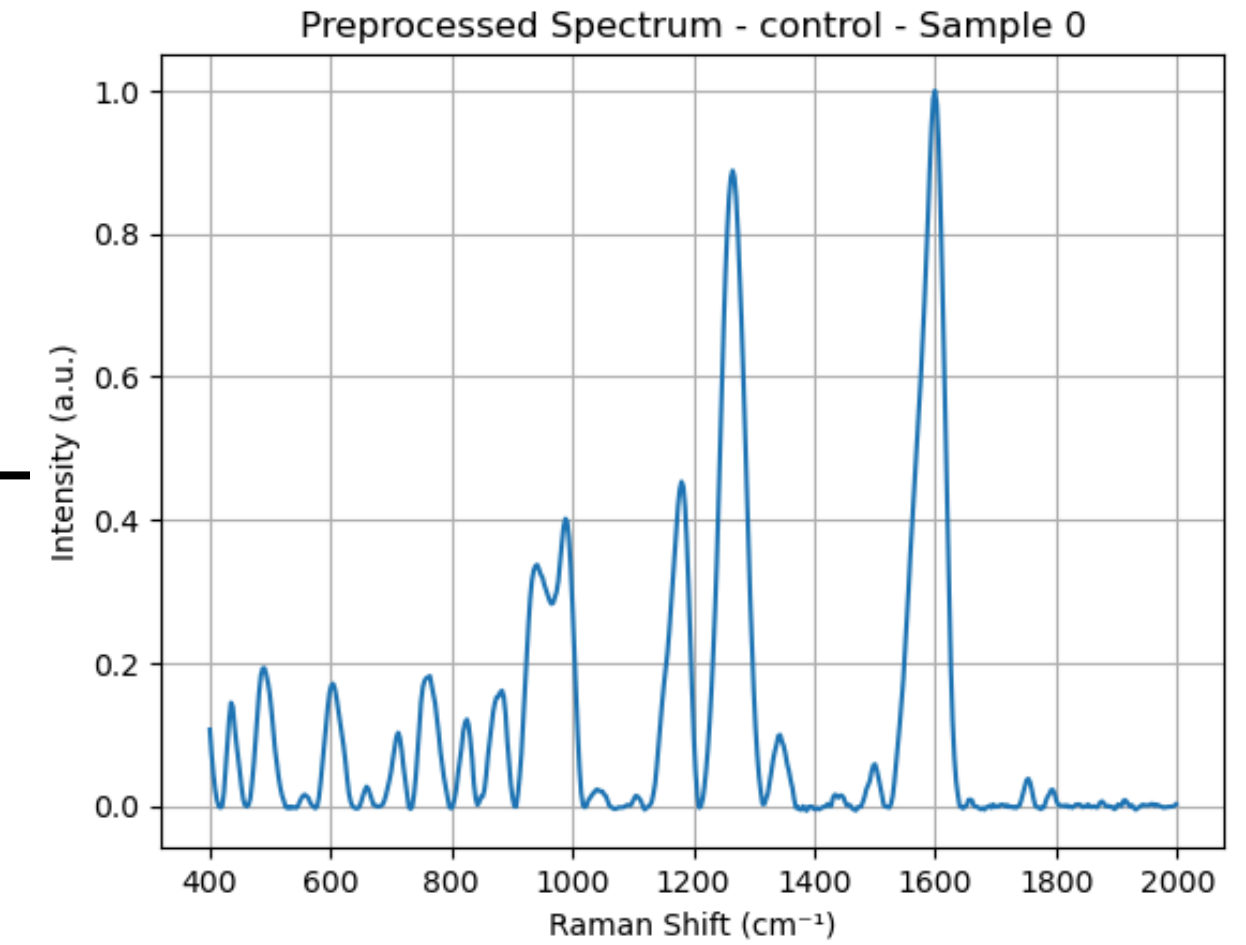
- **Control** - 150
- **Mix** - 145
- **Sulfamethoxazole** - 145
- **Sulfapyridine** - 150
- **Sulfathiazole** - 139

Preprocessing



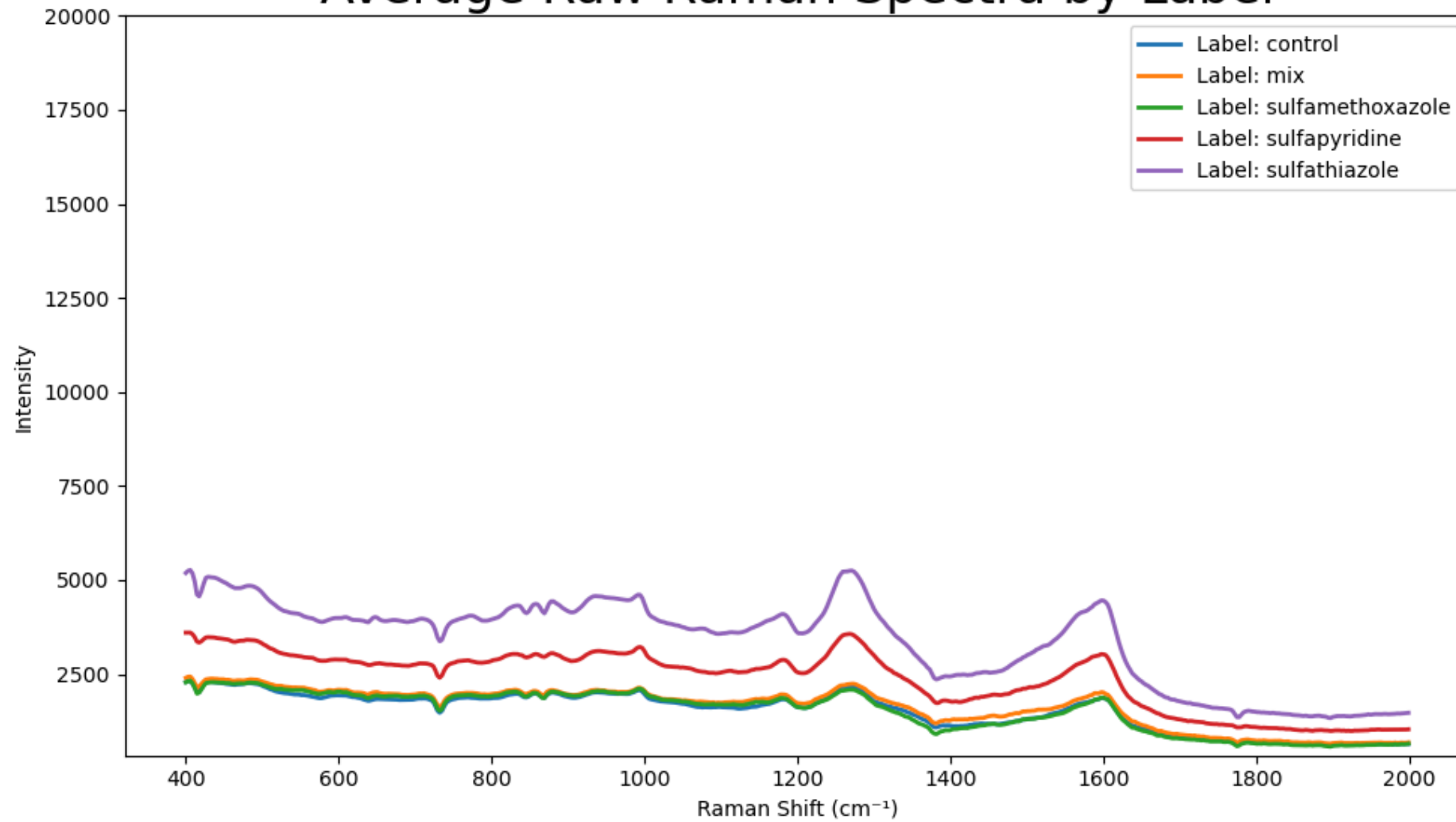
Pipeline

1st Cropping
Cosmic Ray Removal
Denoising
Smoothing
Normalization
2nd Cropping

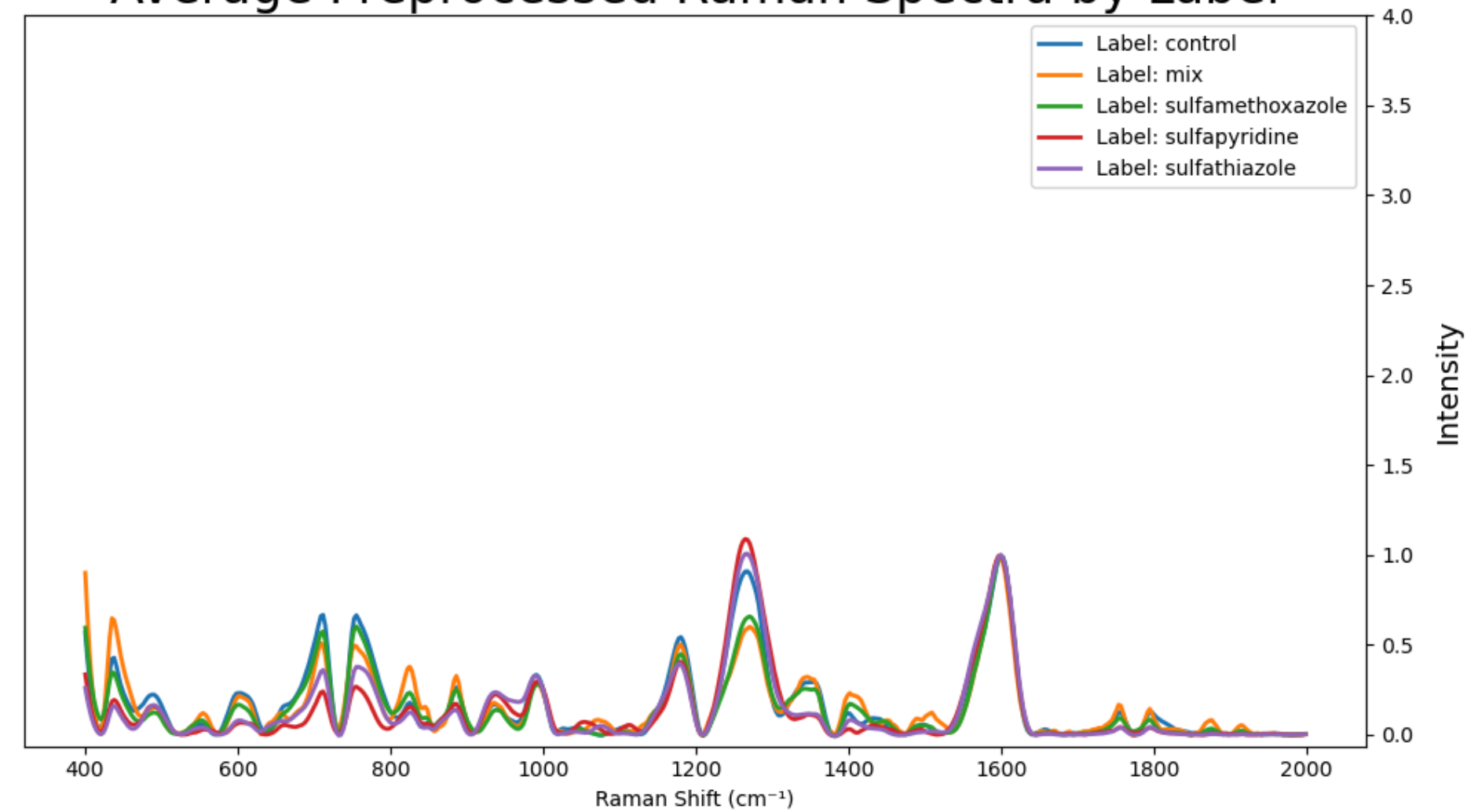


Raw Spectra VS Preprocessed

Average Raw Raman Spectra by Label



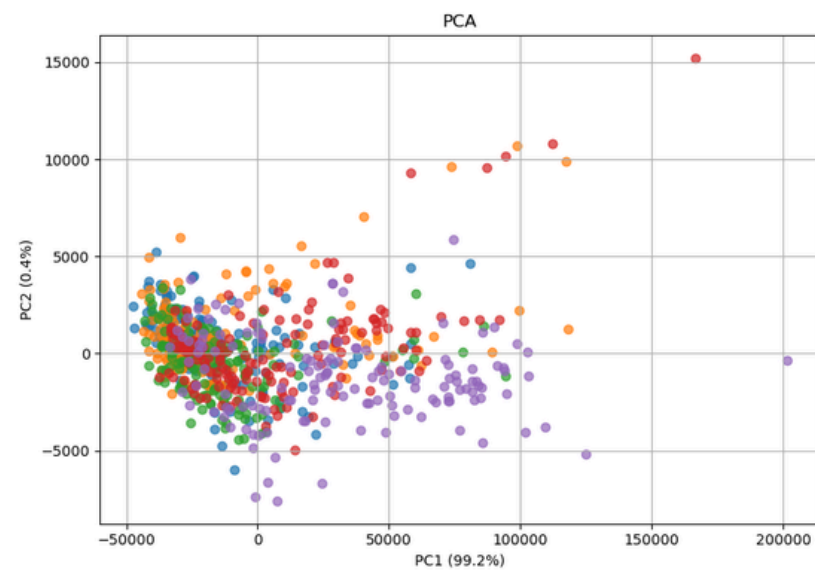
Average Preprocessed Raman Spectra by Label



After applying the preprocessing pipeline to the Raman spectra, a significant improvement in the quality and clarity of the relevant peaks can be observed.

Dimensionality Reduction

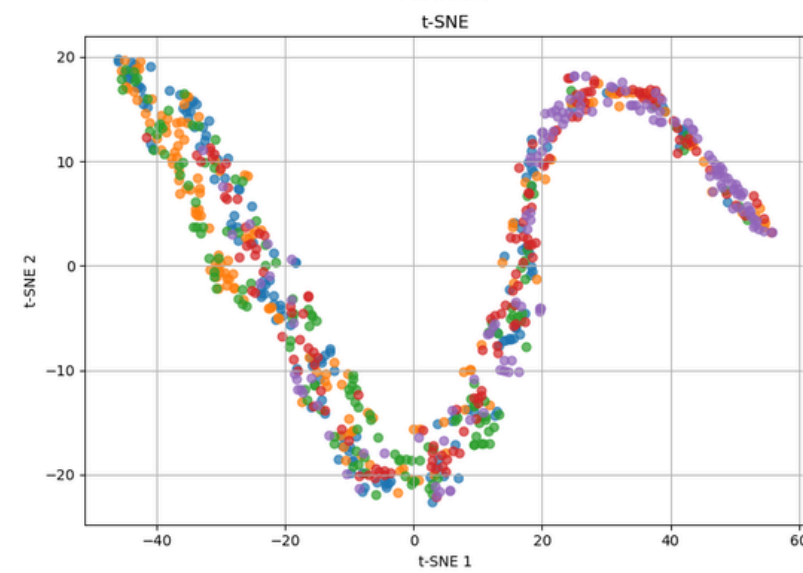
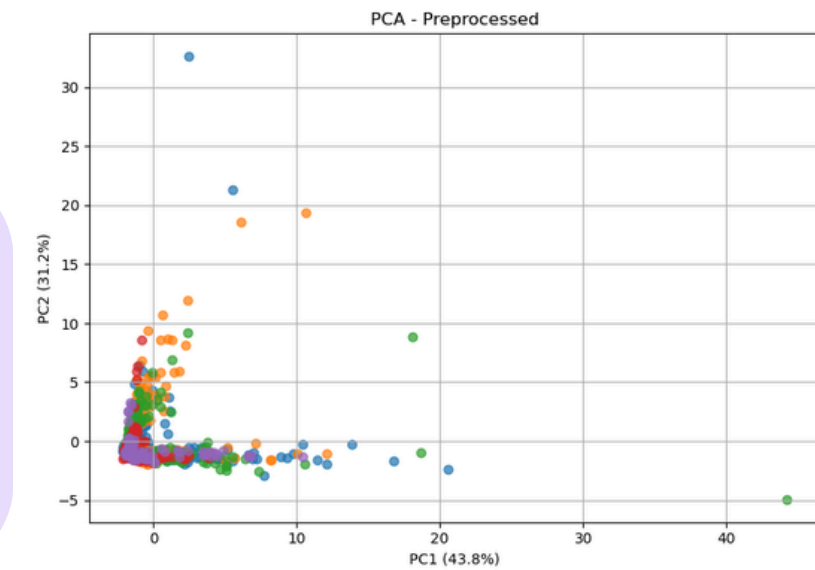
RAW



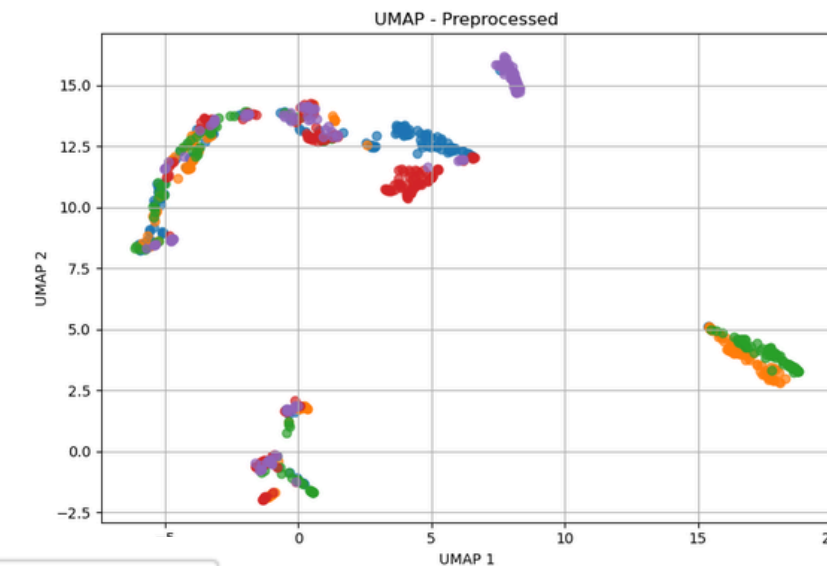
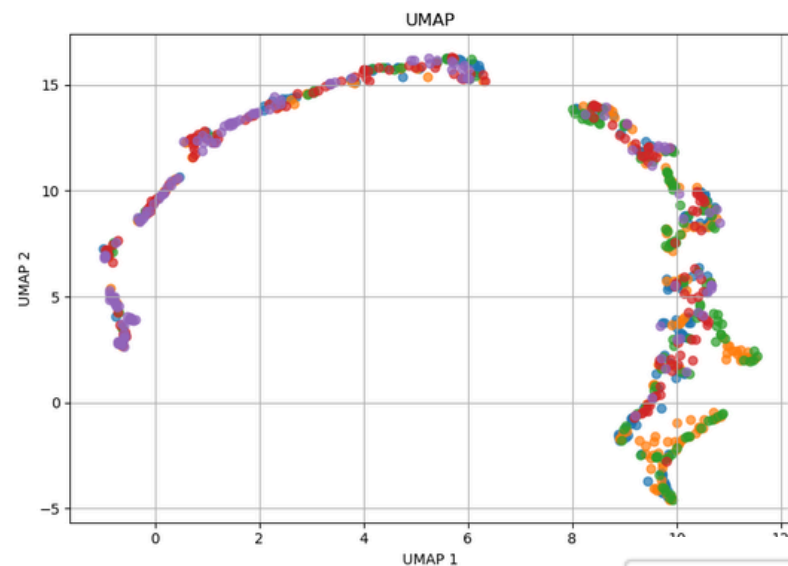
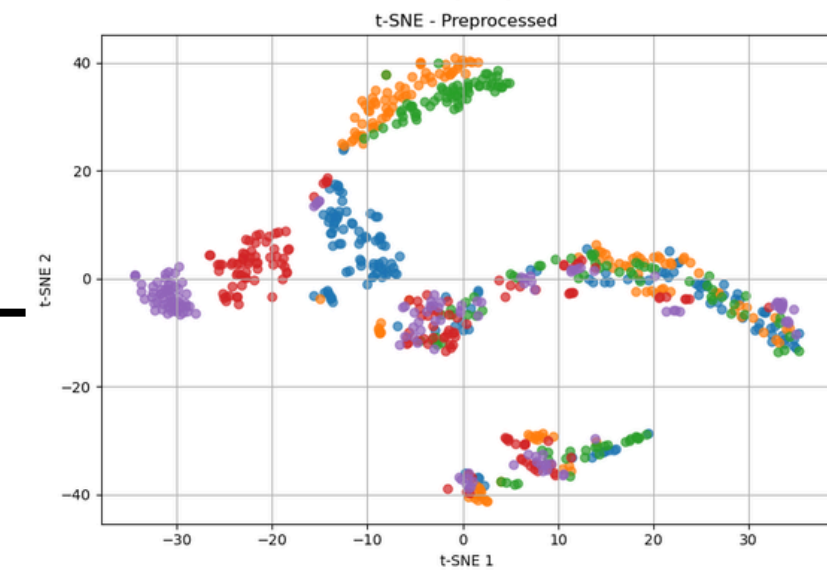
Explained variance
by 2 components:
[0.99208455
0.00377039]

Explained variance
by 2 components:
[0.4381767
0.31204339]

PreProcessed

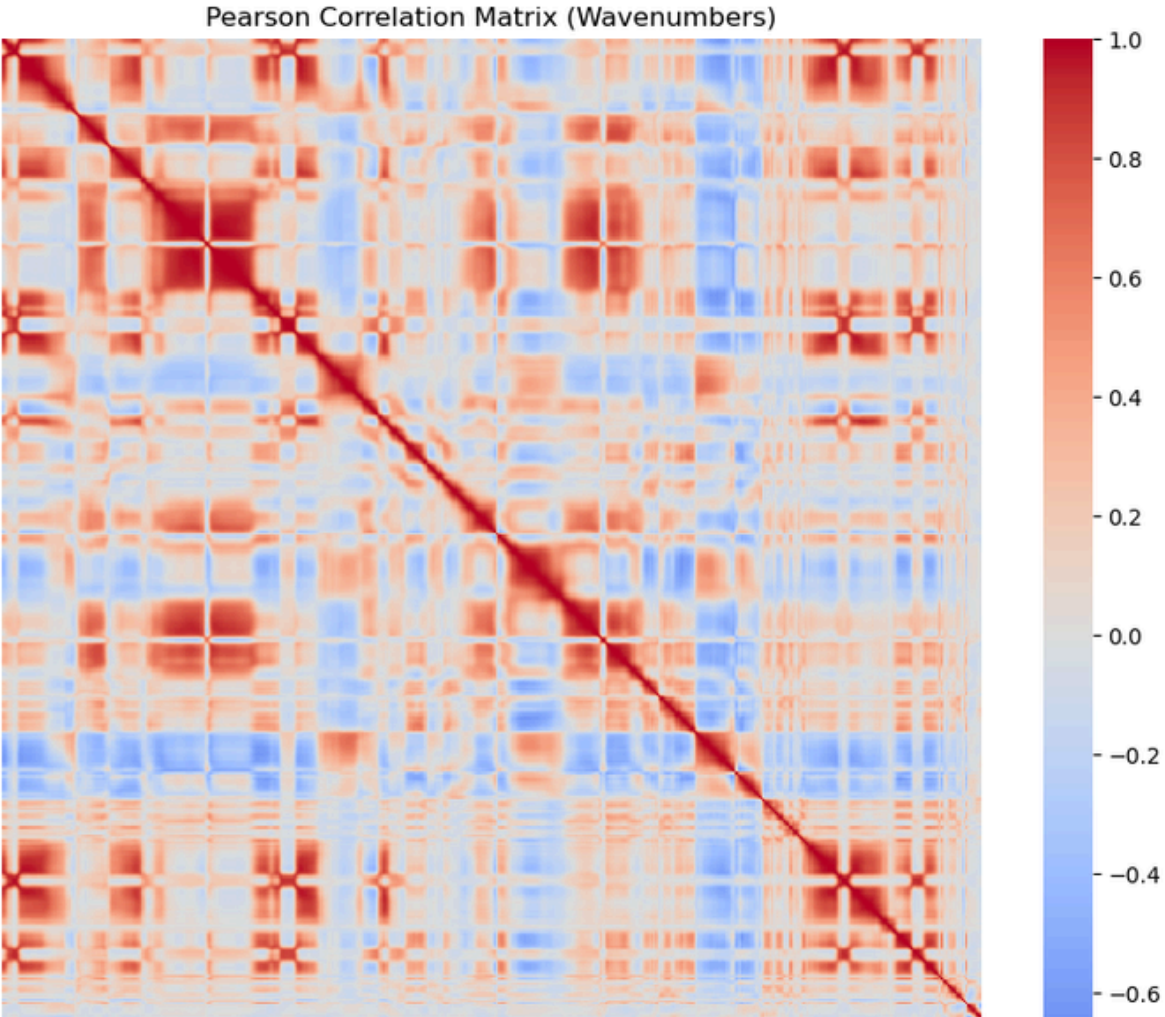


Pipeline

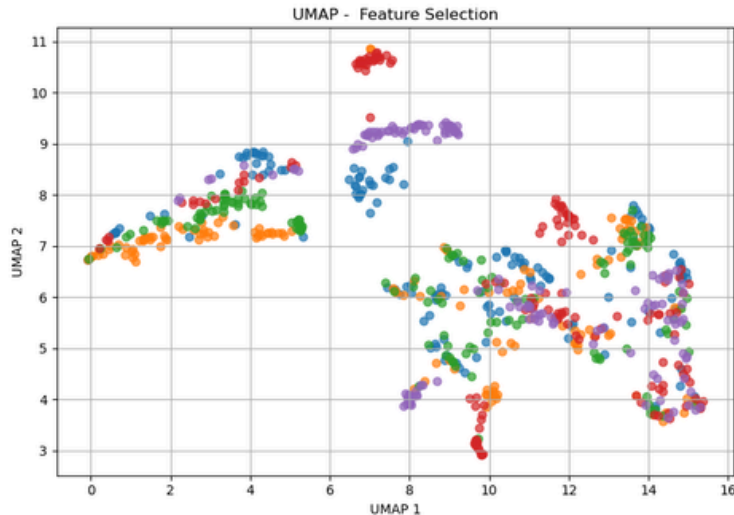
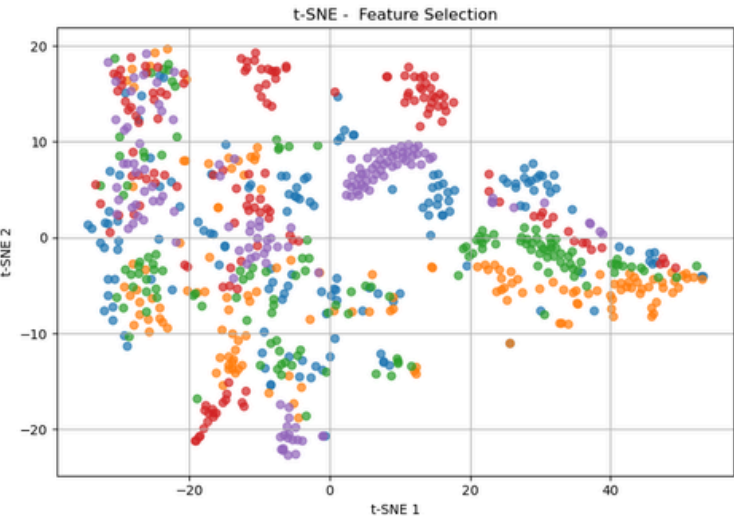
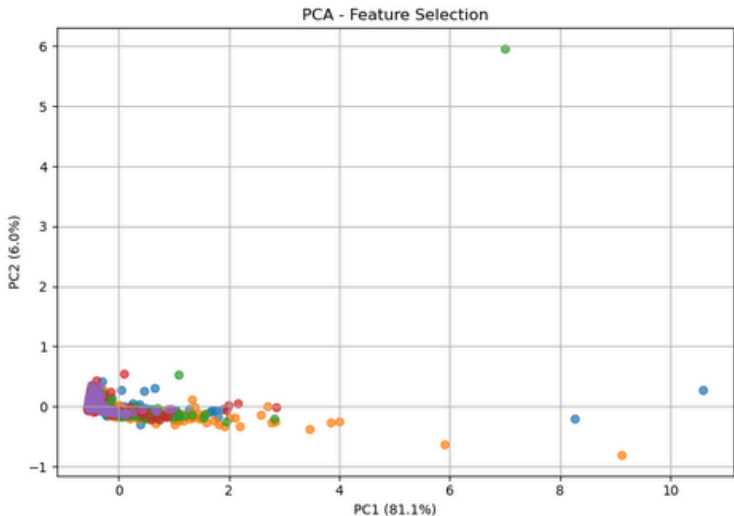


● control ● mix ● sulfamethoxazole ● sulfapyridine ● sulfathiazole

Feature Selection



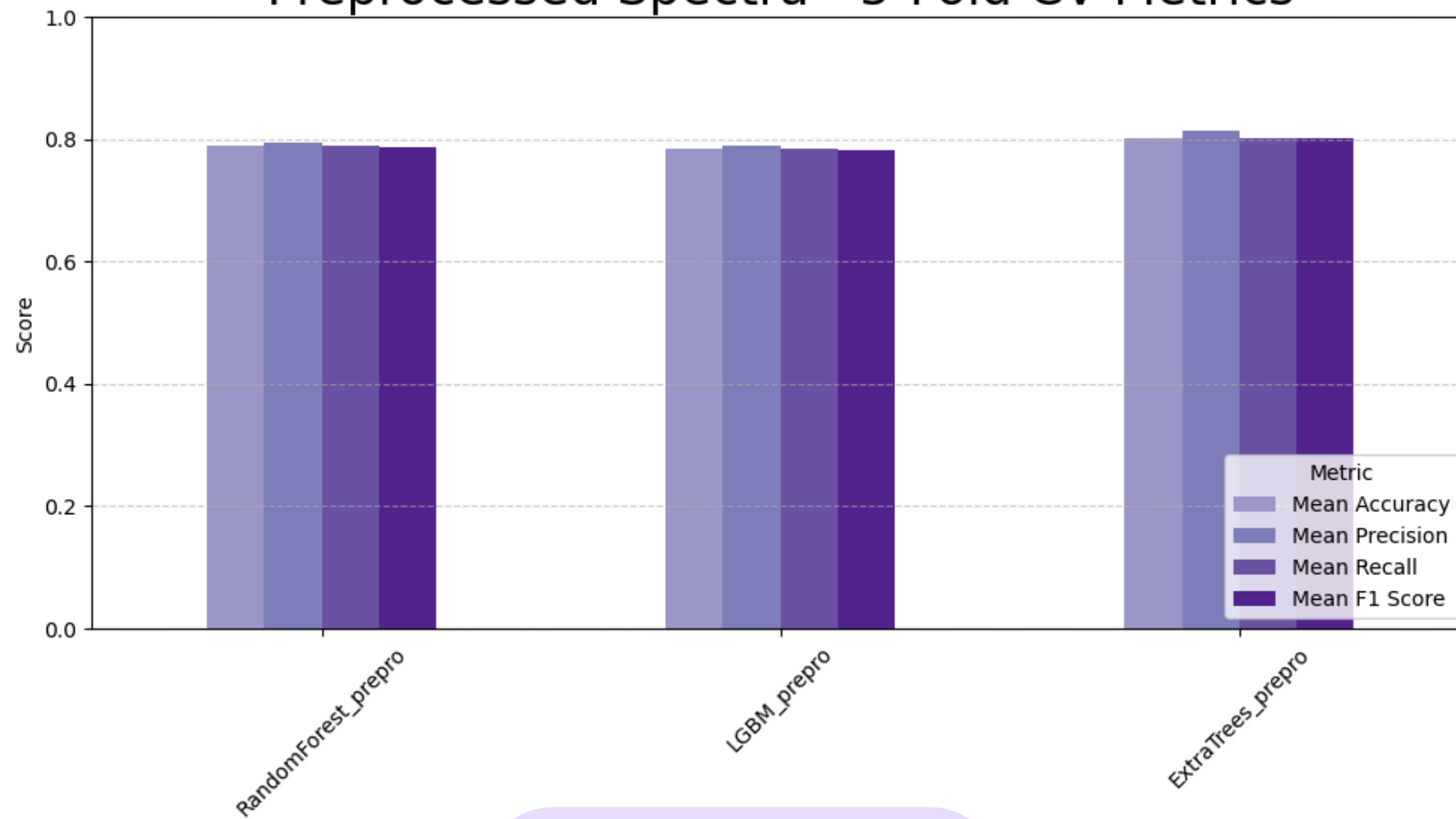
Highly correlated wavenumbers are removed to reduce dimensionality



control mix sulfamethoxazole sulfapyridine sulfathiazole

Machine Learning Exploratory

Preprocessed Spectra - 5-Fold CV Metrics



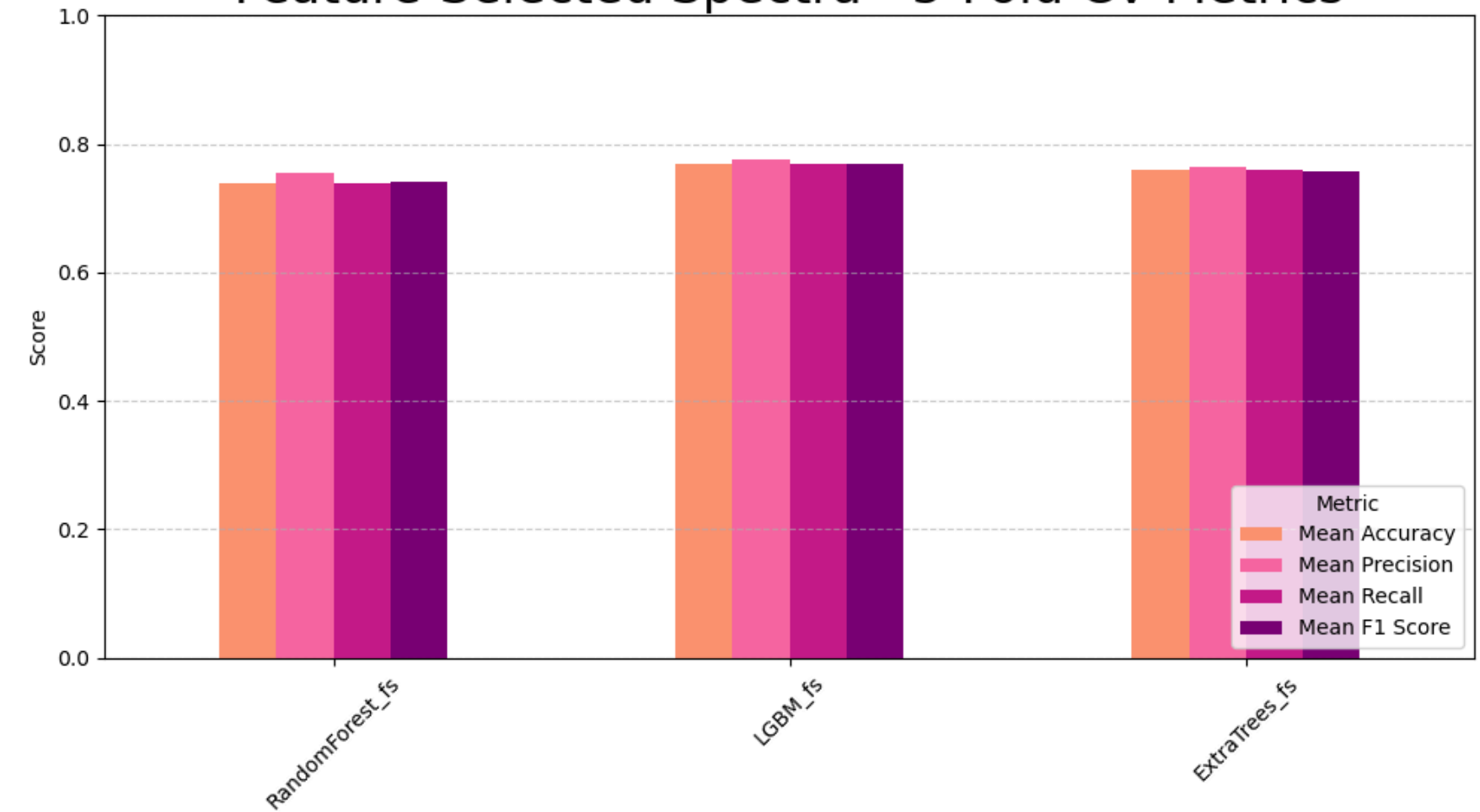
Mean Accuracy

RandomForest - 0.79

LGBM - 0.78

ExtraTrees - 0.80

Feature-Selected Spectra - 5-Fold CV Metrics



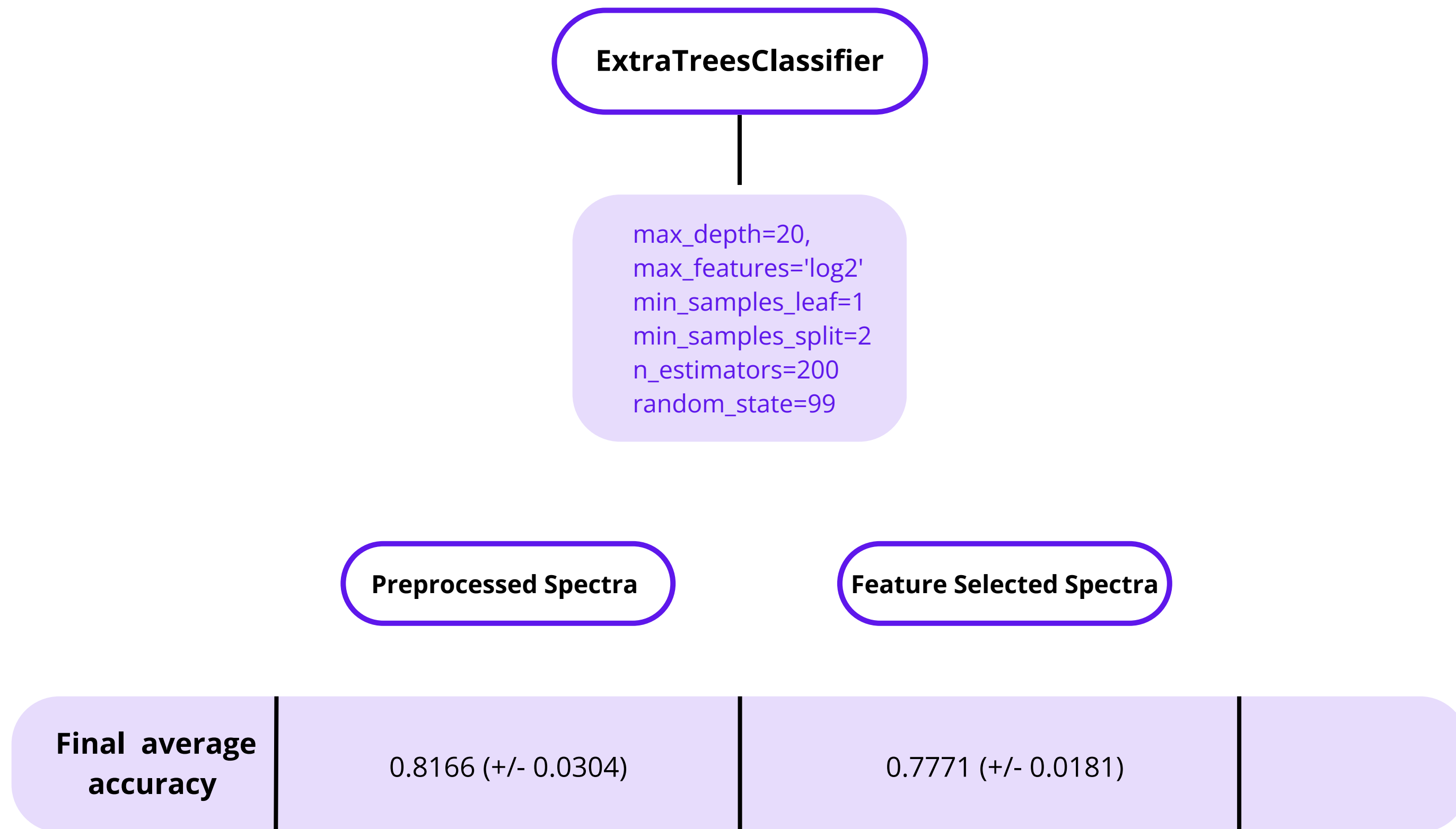
Mean Accuracy

RandomForest - 0.74

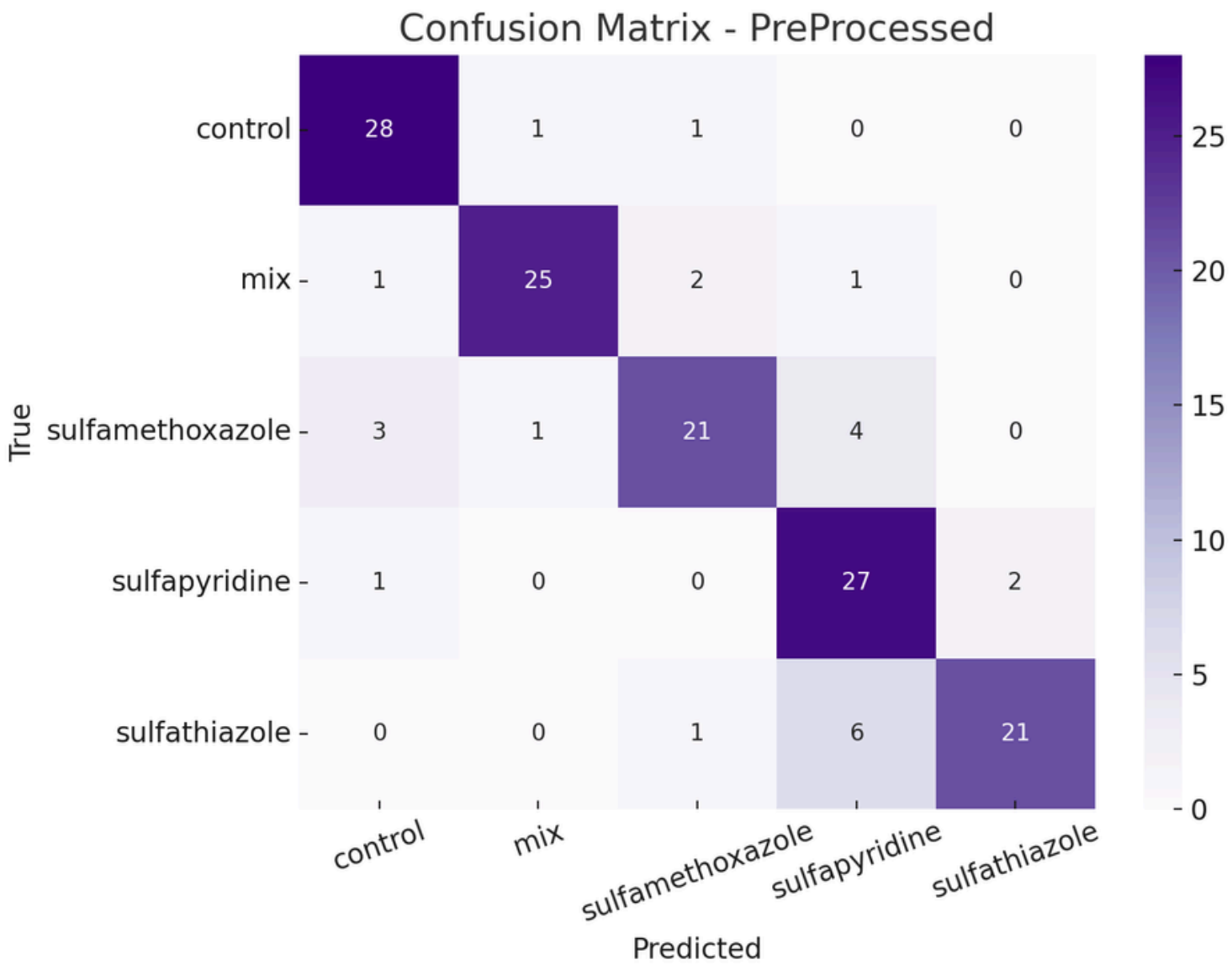
LGBM - 0.77

ExtraTrees - 0.76

Model Fine Tuning

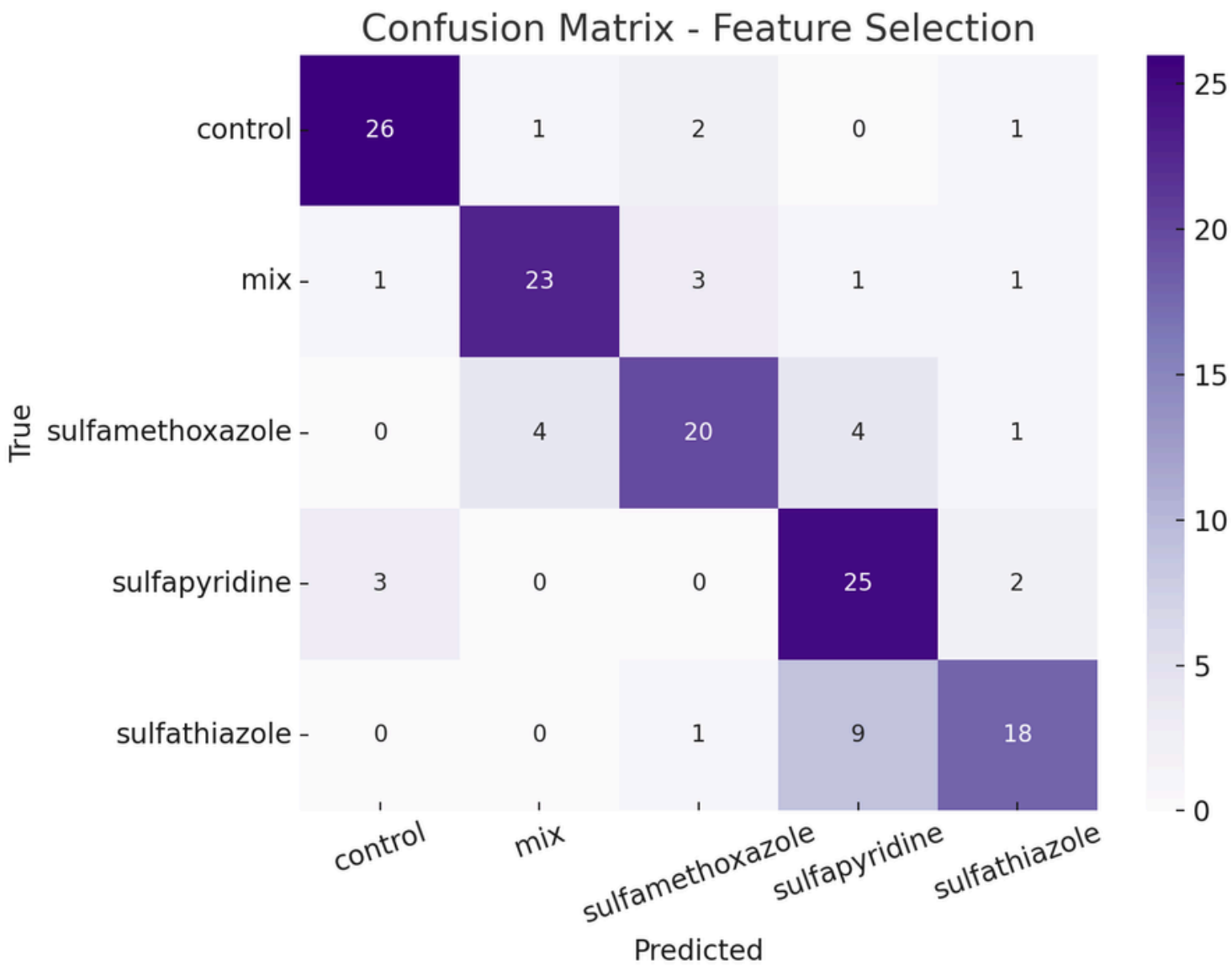


Model Deployment (Test Set 20%)



Classification Report:

	precision	recall	f1-score	support
control	0.85	0.93	0.89	30
mix	0.93	0.86	0.89	29
sulfamethoxazole	0.84	0.72	0.78	29
sulfapyridine	0.71	0.90	0.79	30
sulfathiazole	0.91	0.75	0.82	28
accuracy			0.84	146
macro avg	0.85	0.83	0.84	146
weighted avg	0.85	0.84	0.84	146



Classification Report:

	precision	recall	f1-score	support
control	0.87	0.87	0.87	30
mix	0.82	0.79	0.81	29
sulfamethoxazole	0.77	0.69	0.73	29
sulfapyridine	0.64	0.83	0.72	30
sulfathiazole	0.78	0.64	0.71	28
accuracy			0.77	146
macro avg	0.78	0.77	0.77	146
weighted avg	0.78	0.77	0.77	146

Final remarks

Pre processing pipeline effectiveness

Feature Selection advantages

Future prespective for quantification